IGCSE 0607 Practice Questions — E5.2 — Combined Questions & Markscheme

Angle measurement in degrees

Total Questions: 9 — **Total Marks:** 56 Marks

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- **E5.2** Angle measurement in degrees
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

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1 The positions of three towns, A, B and C, are marked on a map.

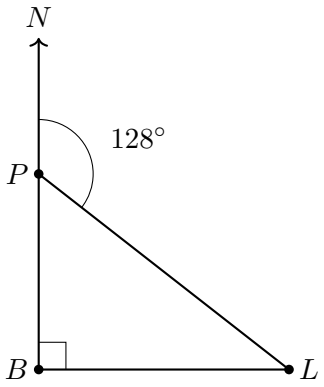
(a) The bearing of town B from town A is 068°.
Work out the bearing of town A from town B.

..... ° **[2]**

(b) Town C is due west of town B.
Write down the bearing of town C from town B.

..... ° **[1]**

2 The bearing of a lighthouse L from a port P is 128°.
A boat B is positioned such that it is due South of P and due West of L.



(a) Find the bearing of P from L.

..... ° **[2]**

(b) Work out the size of angle BPL.

..... ° **[1]**

(c) Work out the size of angle PLB.

..... ° **[2]**

3 The bearing of a lighthouse, L, from a boat, B, is 148°.

(a) Work out the bearing of B from L.

..... ° **[2]**

(b) A port, P, is due north of L and due east of B.

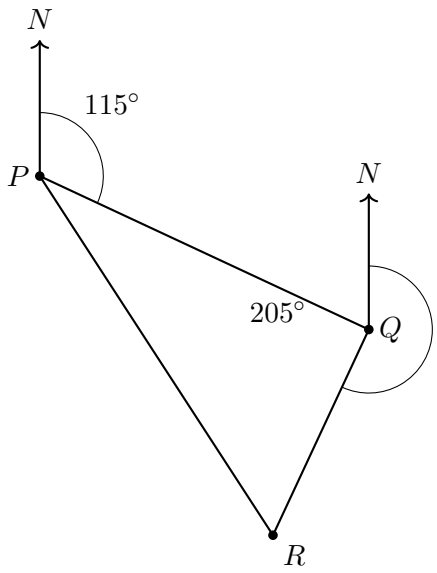
(i) Write down the bearing of B from P.

..... ° **[1]**

(ii) Work out the size of angle LBP.

..... ° **[2]**

4 The diagram shows the positions of three points P, Q and R.
The bearing of Q from P is 115°.
The bearing of R from Q is 205°.



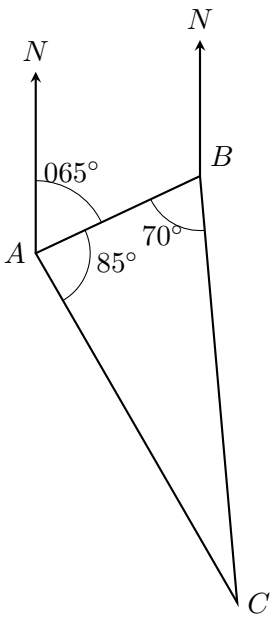
(a) Show that angle $PQR = 90°$.

[2]

(b) The distance PQ is 15 km and the distance QR is 8 km.
Calculate the distance PR.

..... km **[3]**

5 The diagram shows the positions of three towns, A, B and C.
The bearing of B from A is 065°.
Angle $CAB = 85°$ and angle $ABC = 70°$.



(a) Calculate the bearing of C from A.

..... ° **[1]**

(b) Calculate the bearing of A from B.

..... ° **[2]**

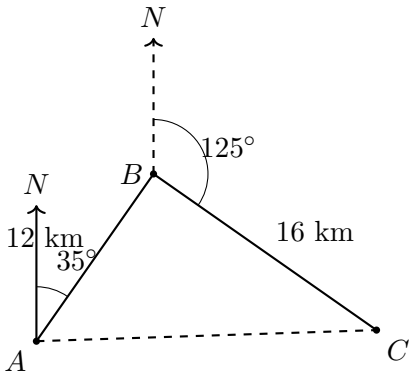
(c) Calculate the bearing of C from B.

..... ° **[2]**

(d) Calculate angle ACB.

..... ° **[1]**

6 A ship leaves a port A and travels 12 km on a bearing of 035° to point B.
From B, the ship travels 16 km on a bearing of 125° to point C.



(a) Show that angle $ABC = 90°$.

[2]

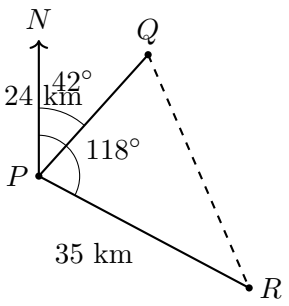
(b) Calculate the direct distance AC.

$AC =$ ..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[3]**

7 Three towns are located at points P, Q, and R.
Town Q is 24 km from P on a bearing of 042°.
Town R is 35 km from P on a bearing of 118°.



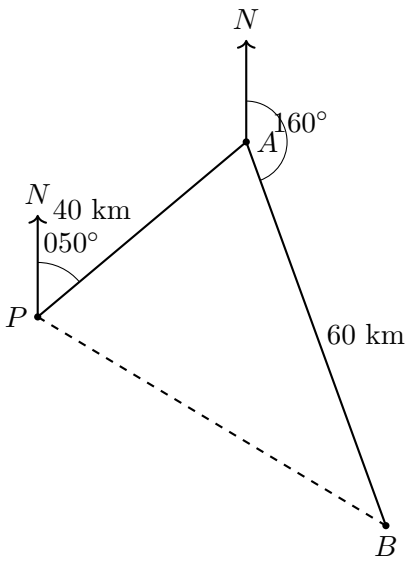
(a) Calculate the distance QR.

$QR =$ ..... km **[3]**

(b) Calculate the bearing of R from Q.

..... ° **[4]**

8 A ship leaves port P and sails 40 km on a bearing of 050° to reach point A.
It then sails 60 km on a bearing of 160° to reach point B.



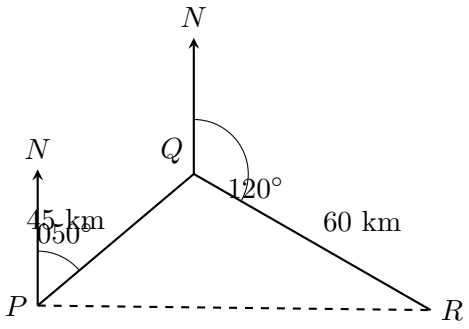
(a) Calculate the distance PB.

..... km **[4]**

(b) Calculate the bearing of B from P.

..... ° **[4]**

9 A ship sails 45 km from Port P on a bearing of 050° to Point Q.
It then sails 60 km on a bearing of 120° to Point R.



(a) Calculate the distance PR.

..... km **[4]**

(b) Calculate the bearing of R from P.

..... ° **[3]**

(c) The ship sails directly back from R to P.
Calculate the bearing of P from R.

..... ° **[2]**

1 **(a)** $068 + 180$ **[M1]**

248 **[A1]**

(b) 270 **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $128 + 180$ **[M1]**

308 **[A1]**

(b) 52 **[B1]**

(c) $180 - 90 - 52$ **[M1]**

38 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $180 + 148$ OR $360 - (180 - 148)$ **[M1]**

328 **[A1]**

(b) **(i)** 270 **[B1]**

(ii) Identifies the bearing of P from B is $090°$ **[B1]**

$148 - 90 = 58$ **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** Bearing of P from $Q = 115 + 180 = 295°$ **[M1]**

Angle $PQR = 295° - 205° = 90°$ **[A1]**

(b) $15^2 + 8^2$ **[M1]**

$225 + 64 = 289$ **[M1]**

17 **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** 150 **[B1]**

(b) $180 + 65$ OR $360 - (180 - 65)$ **[M1]**

245 **[A1]**

(c) Identifies bearing of A from B is $245°$ (can follow through from (b)) **[M1]**

$245 - 70 = 175$ **[A1]**

(d) 25 **[B1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** Interior angle at B (South to BA) $= 35°$ OR interior angle from North line $= 180 - 35 = 145°$ **[M1]**

Angle $ABC = 360 - 145 - 125 = 90°$ OR $125 - 35 = 90°$ **[A1]**

(b) $AC^2 = 12^2 + 16^2$ **[M1]**

$AC^2 = 144 + 256 = 400$ **[M1]**

$AC = 20$ **[A1]**

(c) $\tan(\angle BAC) = \dfrac{16}{12}$ or $\sin(\angle BAC) = \dfrac{16}{20}$ **[M1]**

$\angle BAC = 53.13°$ **[A1]**

Bearing $= 035° + 53.13° = 088.1°$ **[A1]**

[Back to Question] *[Total Marks: 8]*

7 **(a)** $\angle QPR = 118° - 42° = 76°$ **[M1]**

$QR^2 = 24^2 + 35^2 - 2(24)(35)\cos(76°)$ **[M1]**

$QR = 37.3$ or $37.34\ldots$ **[A1]**

(b) $\dfrac{\sin(\angle PQR)}{35} = \dfrac{\sin(76°)}{37.34}$ OR $\cos(\angle PQR) = \dfrac{24^2 + 37.34° - 35^2}{2(24)(37.34)}$ **[M1]**

$\sin(\angle PQR) = 0.909\ldots$ OR $\cos(\angle PQR) = 0.415\ldots$ **[M1]**

$\angle PQR = 65.4°$ or $65.42\ldots°$ **[A1]**

Bearing $= 180 + 42 - 65.4 = 156.6°$ (or $157°$) **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Bearing of P from $A = 050 + 180 = 230°$ **[M1]**

Angle $PAB = 230° - 160° = 70°$ **[M1]**

$PB^2 = 40^2 + 60^2 - 2(40)(60)\cos(70°)$ **[M1]**

59.6 or $59.65\ldots$ **[A1]**

(b) Uses sine rule or cosine rule to find angle APB **[M1]**

$\dfrac{\sin(APB)}{60} = \dfrac{\sin(70°)}{59.65}$ or $\cos(APB) = \dfrac{40^2 + 59.65^2 - 60^2}{2(40)(59.65)}$ **[M1]**

Angle $APB = 70.9°$ $(70.93\ldots°)$ **[A1]**

Bearing $= 050 + 70.9 = 120.9°$ **[A1]**

[Back to Question] *[Total Marks: 8]*

9 **(a)** Identifies angle $PQR = 230° - 120° = 110°$ (or uses parallel lines to find $180 - 50 = 130$ and $360 - 130 - 120 = 110$) **[B1]**

Uses Cosine rule: $PR^2 = 45^2 + 60^2 - 2(45)(60)\cos(110°)$ [M2] (M1 for correct substitution into formula)

$PR = 86.4$ or $86.44\ldots$ **[A1]**

(b) Uses Sine rule: $\dfrac{\sin(QPR)}{60} = \dfrac{\sin(110°)}{86.44}$ **[M1]**

Angle $QPR = 40.7°$ $(40.73\ldots)$ **[A1]**

Bearing $= 050° + 40.7° = 090.7°$ **[A1]**

(c) $090.7 + 180$ **[M1]**

270.7 **[A1]**

[Back to Question] *[Total Marks: 9]*